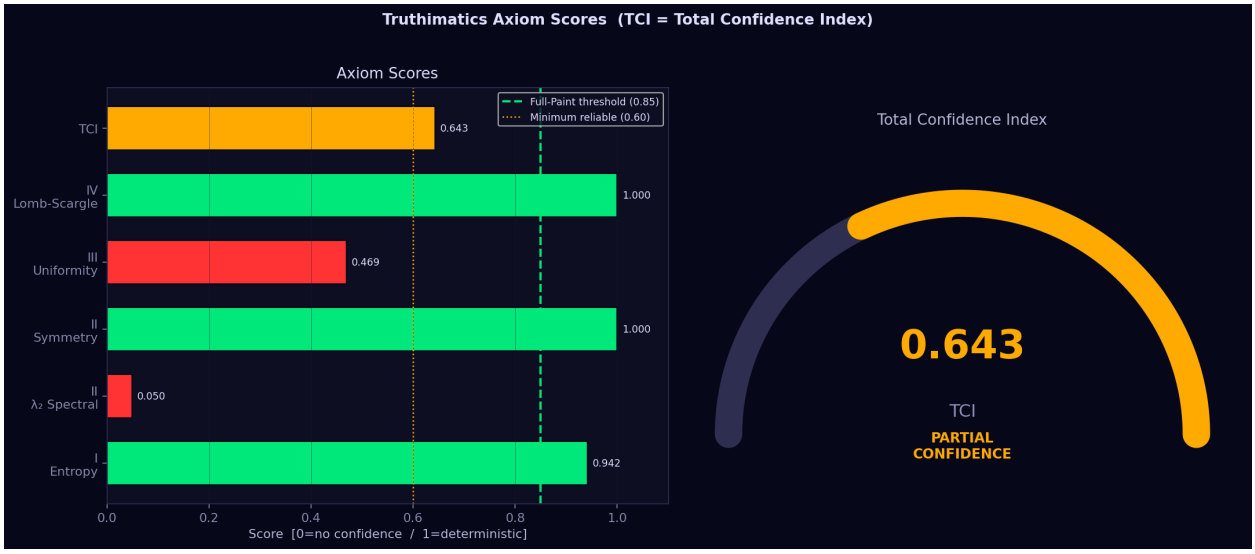
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_CC2BB5A7D3
Taxa	Utahraptor
Clade	Theropod
PBDB Records	4
Mass (regression)	3602.7 kg (Campione & Evans 2012)
Mass (volumetric)	1925.0 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	2.265 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6428 (partial)
Axiom I Entropy	0.9415
Axiom II λ■	0.1992 (symmetry 100%)
Axiom III Uniformity	0.4690
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	68.0% Measured / 32.0% Inferred
Elements	25
CoG (m)	[0.265, 0.497, 0.0]
Generated	2026-04-30 23:53 UTC

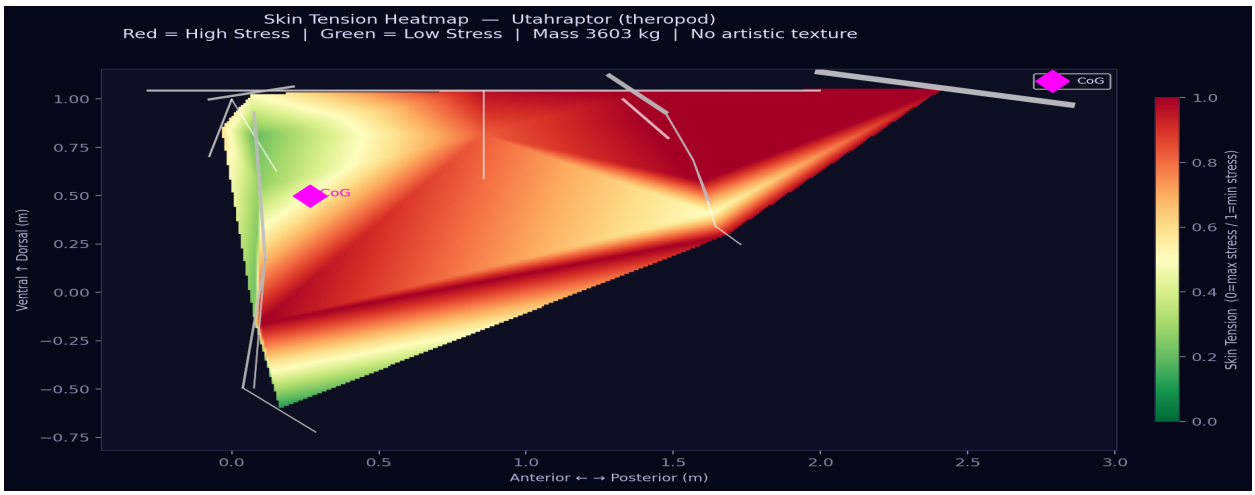
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



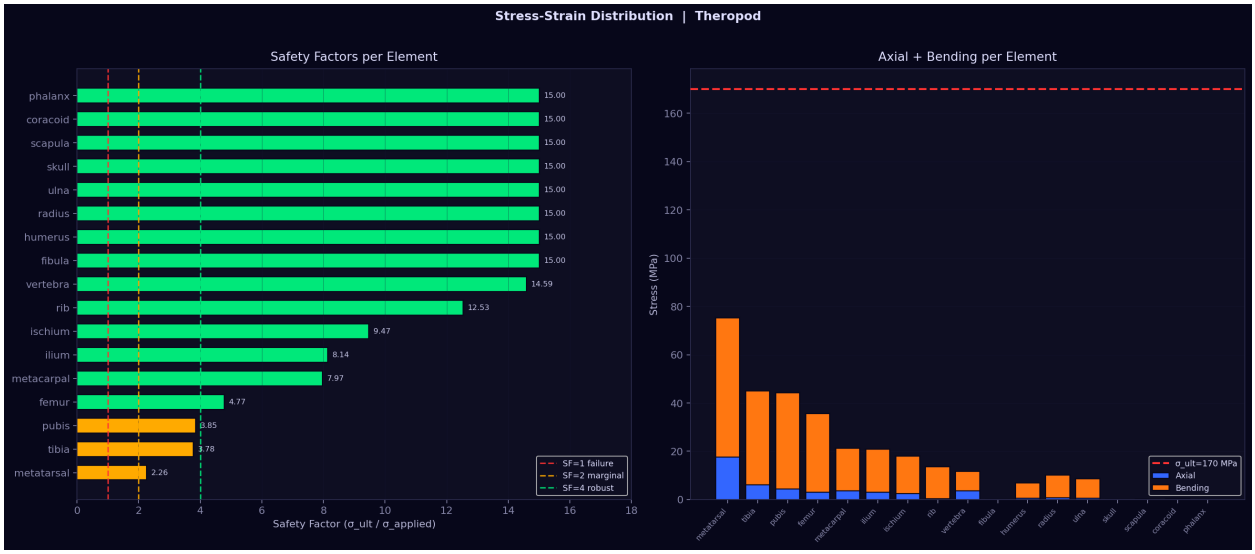
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

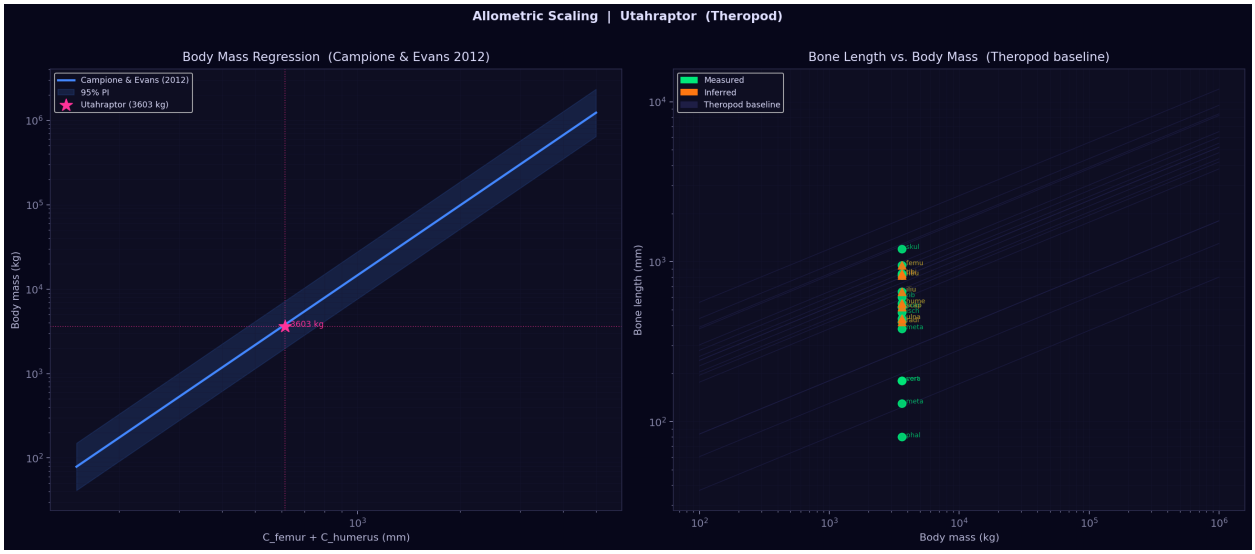


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.086	0.173	0.259	15.000	OK
femur	3.064	32.556	35.620	4.773	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.627	6.176	6.804	15.000	OK
ilium	3.064	17.820	20.884	8.140	OK
ischium	2.413	15.545	17.957	9.467	OK
metacarpal	3.684	17.641	21.325	7.972	OK
metatarsal	17.647	57.607	75.254	2.259	WARNING
phalanx	0.000	0.000	0.000	15.000	OK
pubis	4.289	39.917	44.206	3.846	WARNING
radius	0.700	9.400	10.100	15.000	OK
rib	0.408	13.159	13.568	12.530	OK
scapula	0.031	0.108	0.139	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	6.106	38.879	44.985	3.779	WARNING
ulna	0.610	8.006	8.616	15.000	OK
vertebra	3.646	8.008	11.654	14.587	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.17 m	Femur + tibia + metatarsal
Estimated Speed	16.8 km/h (4.67 m/s)	Gait: Trot | Fr=1.024
Stride Length	5.012 m	Alexander (1976) / Thulborn (1990)
Bite Force	24473.9 N (2494.8 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	1200.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3968	Circumference / length (dimensionless)
Limb Strength Index	3.6e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	8.1 m	Allometric from femur length
Est. Neck Length	1.13 m	Allometric from femur length
Est. Tail Length	3.08 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	180.0	120.0	1038.23	1296.0	Measured	0.88	15.00
femur	950.0	120.0	5479.57	6840.1	Measured	0.88	4.77
femur_right	950.0	120.0	5479.57	6840.1	Symmetry	0.84	4.77
fibula	820.0	35.0	284.02	297.9	Measured	0.88	15.00
fibula_right	820.0	35.0	284.02	297.9	Symmetry	0.84	15.00
humerus	550.0	75.0	1239.21	1546.9	Measured	0.88	15.00
humerus_right	550.0	75.0	1239.21	1546.9	Symmetry	0.84	15.00
ilium	650.0	120.0	3749.18	4680.1	Measured	0.88	8.14
ilium_right	650.0	120.0	3749.18	4680.1	Symmetry	0.84	8.14
ischium	480.0	80.0	1230.50	1536.0	Measured	0.88	9.47
metacarpal	130.0	30.0	17.46	14.4	Measured	0.88	7.97
metatarsal	380.0	50.0	380.53	475.0	Measured	0.88	2.26
phalanx	80.0	25.0	6.03	4.7	Measured	0.88	15.00
pubis	520.0	60.0	749.84	936.0	Measured	0.88	3.85
radius	420.0	42.0	296.76	370.4	Measured	0.88	15.00
radius_right	420.0	42.0	296.76	370.4	Symmetry	0.84	15.00
rib	600.0	30.0	216.30	270.0	Measured	0.88	12.53
scapula	520.0	200.0	8331.50	10400.2	Measured	0.88	15.00
scapula_right	520.0	200.0	8331.50	10400.2	Symmetry	0.84	15.00
skull	1200.0	500.0	120165.92	150003.1	Measured	0.88	15.00
tibia	840.0	85.0	2430.96	3034.6	Measured	0.88	3.78
tibia_right	840.0	85.0	2430.96	3034.6	Symmetry	0.84	3.78
ulna	440.0	45.0	356.89	445.5	Measured	0.88	15.00
ulna_right	440.0	45.0	356.89	445.5	Symmetry	0.84	15.00
vertebra	180.0	110.0	872.40	1089.0	Measured	0.88	14.59